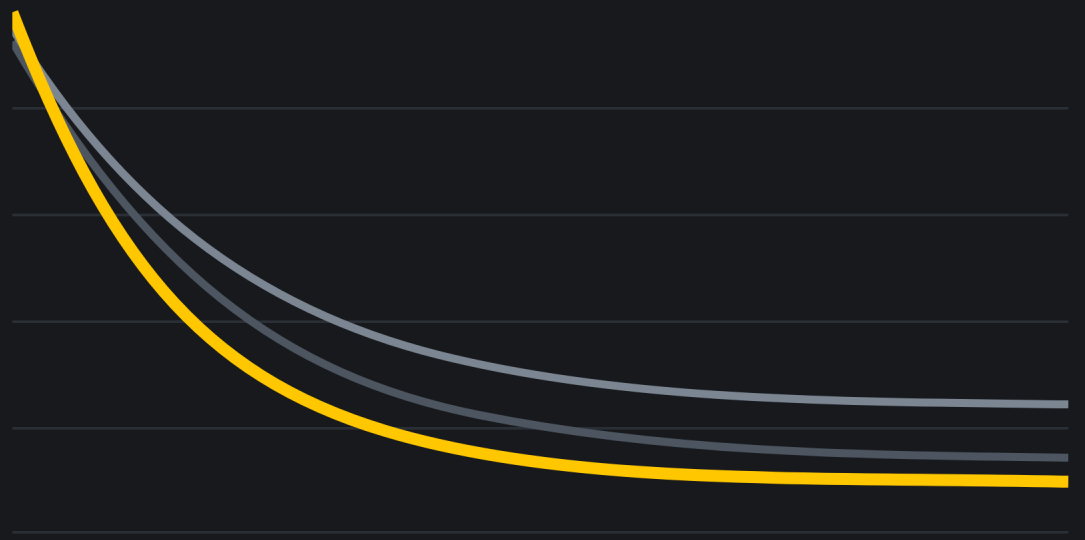


Weights & Biases

Experiment tracking for machine learning

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After a few dozen runs, nobody knows which settings made the best model

You need	Without tracking	With W&B
Settings	File names, notebooks	Saved as config
Metrics	Screenshots	Live charts
Code	Whatever was on disk	Git commit recorded
Model	A .pt file	Versioned and linked

W&B records every step of training. It never trains the model itself.

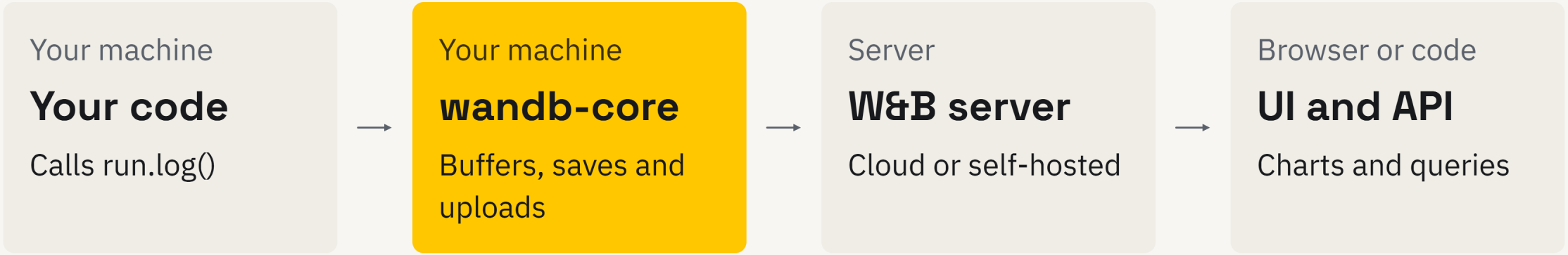
Data	**Artifacts**	Datasets get versions
Train	**Runs**	Metrics, config and system stats
Tune	**Sweeps**	Grid, random or Bayesian search
Evaluate	**Tables**	Browse predictions and mistakes
Deploy	**Registry**	Mark a model version for production

You keep PyTorch, Keras or whatever you use.

Six objects cover most of W&B

Object	What it is	In code
Run	One run of your training code	wandb.init()
Config	Inputs, like learning rate	run.config
History	Metrics logged at each step	run.log()
Project	Runs you want to compare	project="digits"
Artifact	A versioned dataset or model	wandb.Artifact()
Sweep	A hyperparameter search	wandb.sweep()

Logging runs in the background, so training never waits



No internet? Set WANDB_MODE=offline, then run wandb sync later.

Works with any Python code. PyTorch, Keras and Hugging Face have ready integrations.

Seven extra lines add tracking

```
import wandb
with wandb.init(project="digits", config=cfg) as run:
    model = SmallCNN(run.config.channels)
    run.watch(model)
    for epoch in range(run.config.epochs):
        acc, loss = train_one_epoch(model)
        run.log({"val/acc": acc, "val/loss": loss})
    art = wandb.Artifact("digits-cnn", type="model")
    art.add_file("model.pt")
    run.log_artifact(art)
```

Yellow lines are W&B. The rest is your normal training loop.

`wandb.init`

Starts a run and saves the config

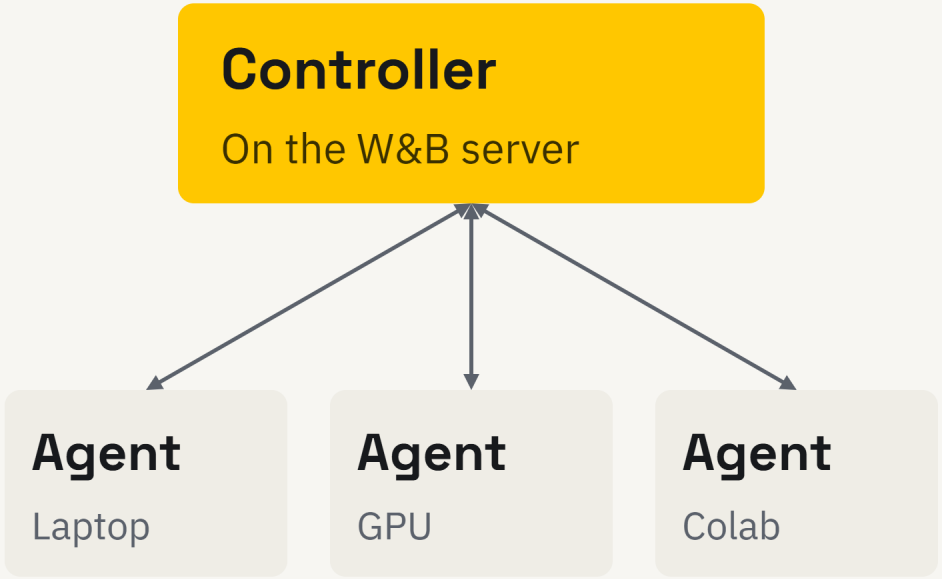
`run.log`

Sends metrics, images and tables

`log_artifact`

Saves the model as a new version

The server picks the next settings, and agents run them



Search: grid, random or Bayesian

Early stop: Hyperband ends weak runs

Scale: start agents on any machine

```
sweep_id = wandb.sweep(sweep_config)
wandb.agent(sweep_id, sweep_run, count=10)
```

Artifacts record which run made a model and which run used it



Same file twice? Still one version.

Ask for digits-cnn:latest, not a file path.